

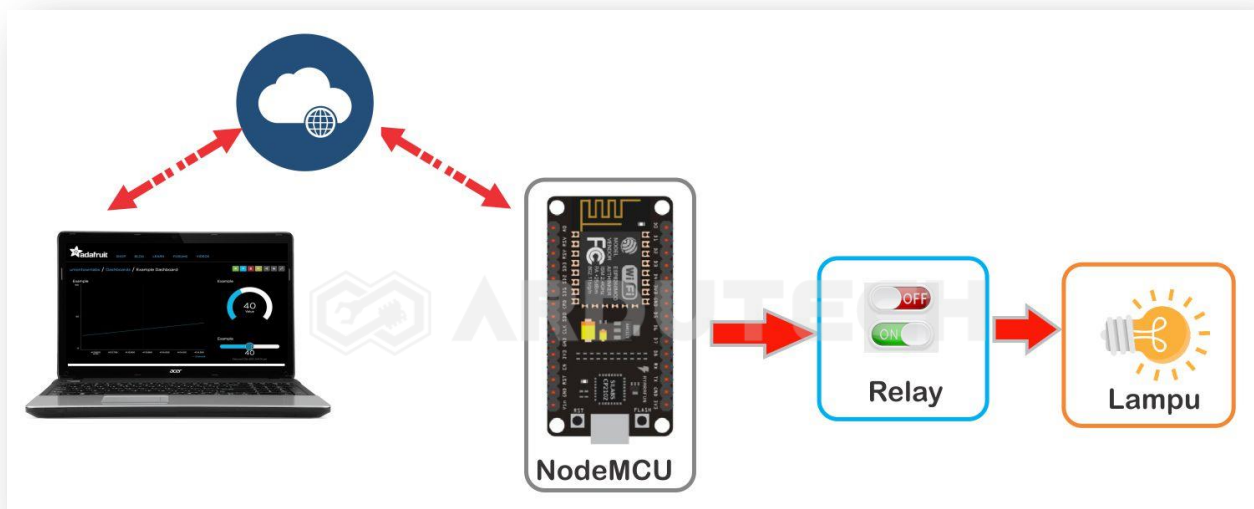
Kontrol 4 Lampu dengan Adafruit IO

Deskripsi

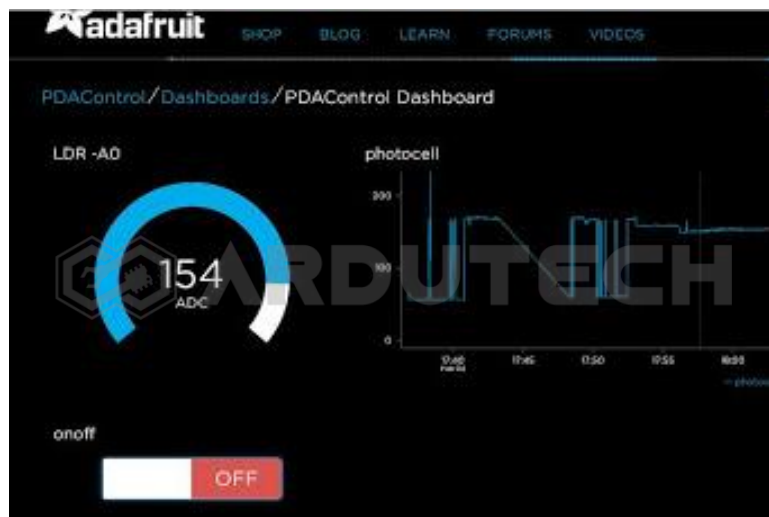
Kontrol On – Off LED melalui aplikasi web io.adafruit.com. Sebagai controllernya NodeMCU ESP8266 yang dihubungkan dengan sebuah lampu LED.

Cara Kerja

NodeMCU ESP8266 memanfaatkan layanan MQTT server yang disediakan oleh Adafruit IO akan memproses command yang diterima kemudian diterjemahkan menjadi sinyal kontrol ON – OFF untuk mengendalikan nyala – padam sebuah lampu/LED.



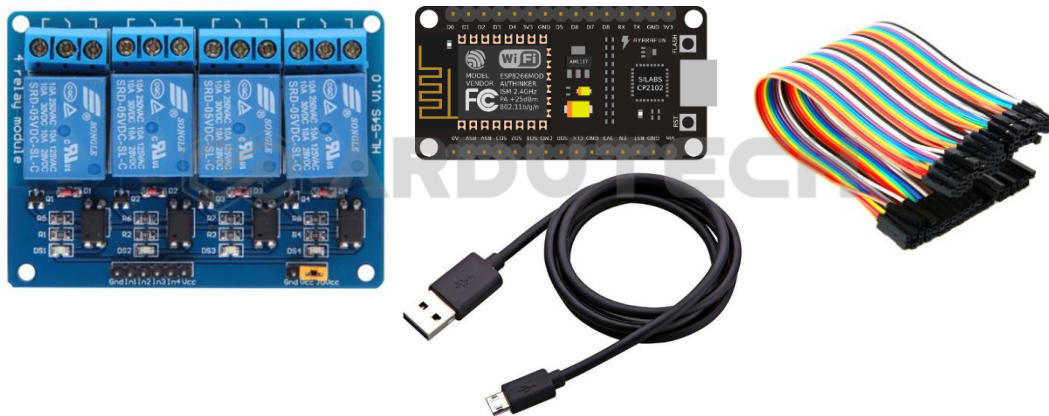
Adafruit IO adalah sistem yang disediakan oleh Adafruit (pembuat modul – modul sensor, perangkat elektronik, robot dll) untuk membuat sebuah proyek/aplikasi IoT (Internet of Things) . Adafruit IO ini boleh dipakai oleh siapa saja karena sifatnya free.



Kebutuhan Bahan

- NodeMCU ESP8266 V3
- Relay modul 4 channel

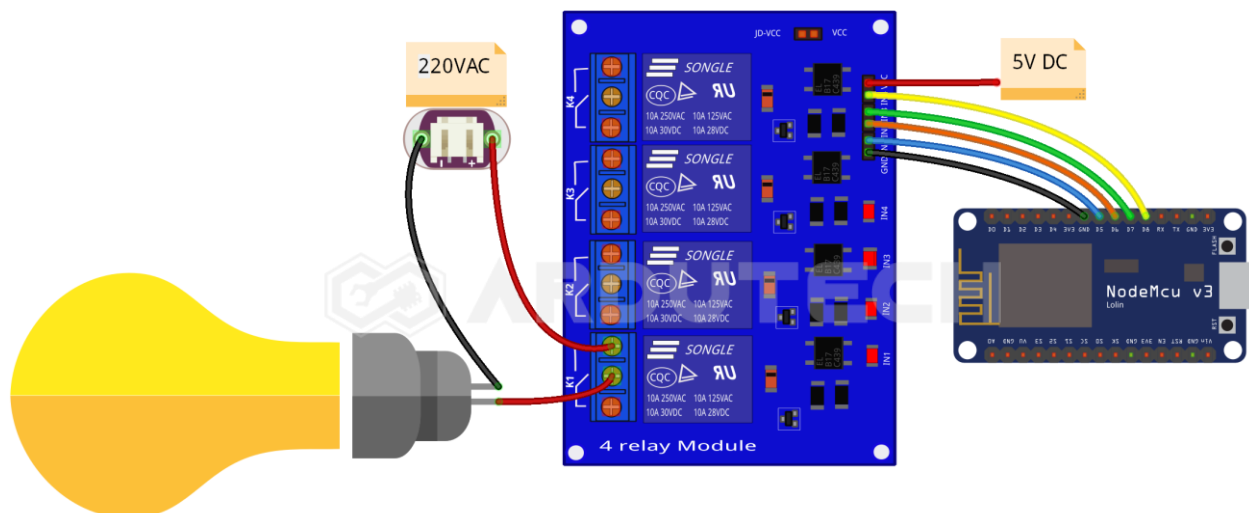
- Kabel micro USB
- Kabel konektor



Kebutuhan Software

- Arduino IDE
- Adafruit IO

Rangkaian/Skematik

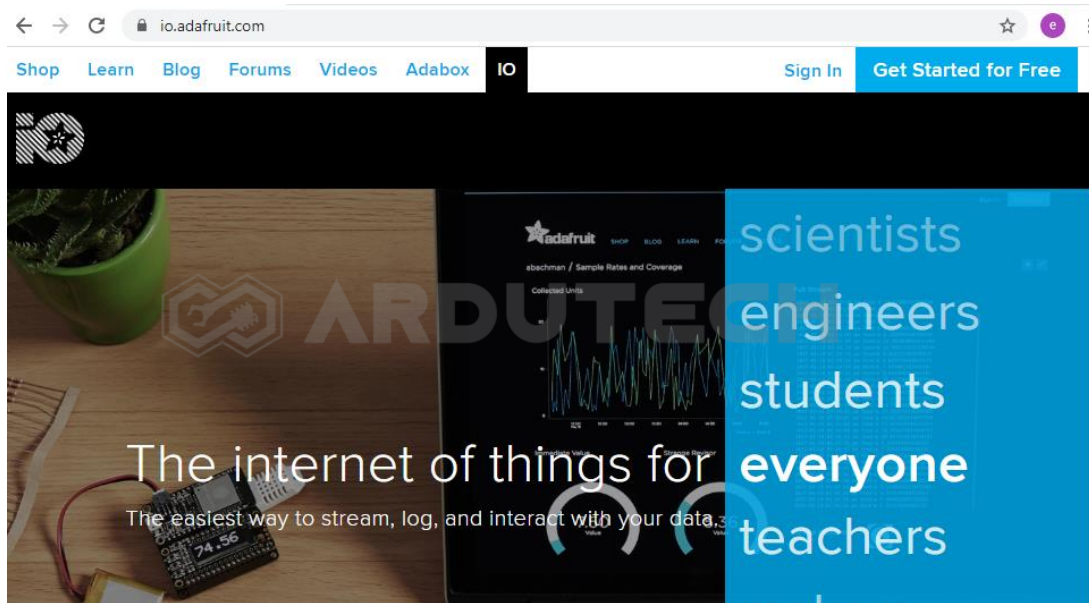


Pada rangkaian hanya ada 1 lampu AC, untuk 3 lampu AC yang lain anda tinggal menambahkan sama dengan lampu 1.

NodeMCU V3	4 Ch Relay Module
D5	IN1
D6	IN2
D7	IN3
D8	IN4

Membuat Antarmuka di Adafruit

Silakan masuk ke halaman <https://io.adafruit.com/>

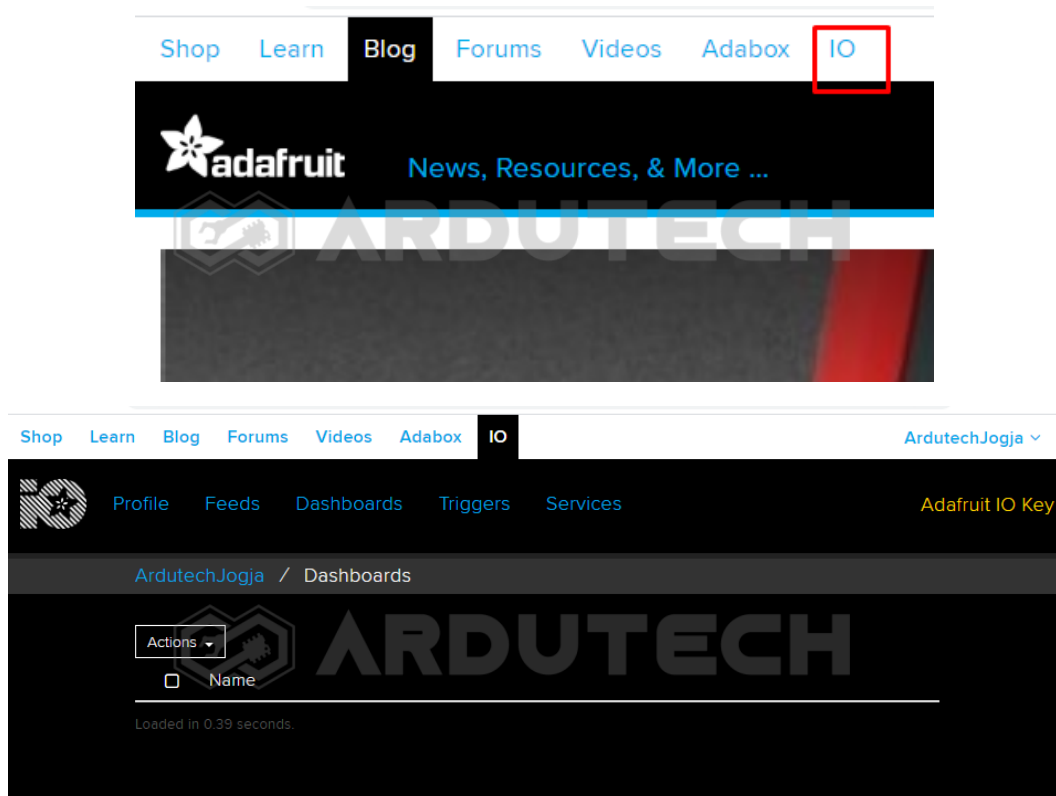


Jika belum mempunyai akun silakan membuat akun terlebih dahulu. Siapkan sebuah akun email aktif.

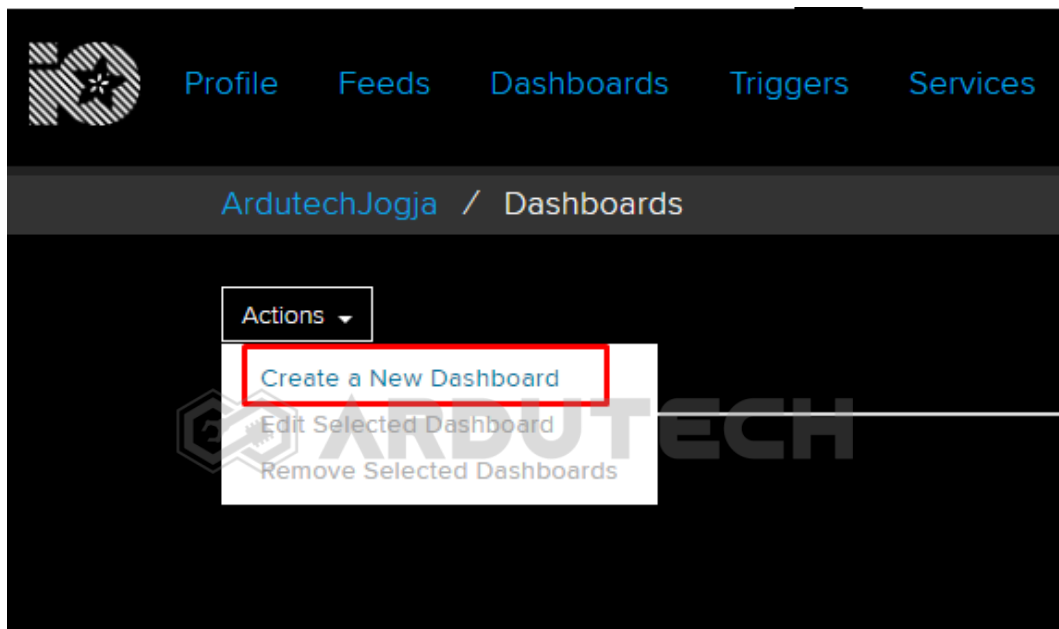
Klik pada **“Get Started for free”** di bagian pojok kanan atas.

Isi data – data yang diperlukan kemudian klik **“CREATE ACCOUNT”** dan selanjutnya ikuti petunjuknya.

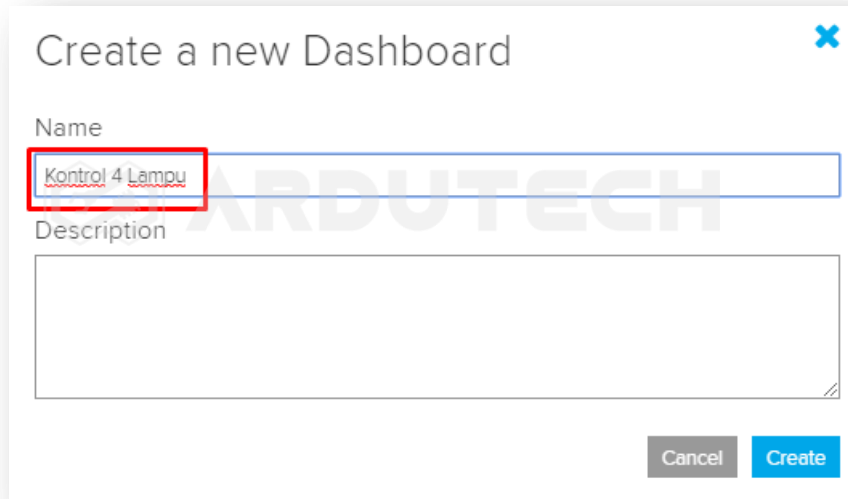
Masuk ke menu **IO**.



Klik pada tombol “Actions” kemudian pilih “Create a New Dashboard”



Selanjutnya akan muncul jendela untuk memberi nama, silakan beri nama pada dashboard yang akan dibuat, misalnya “Kontrol 4 Lampu” dan klik “Create”.



Create a new Dashboard

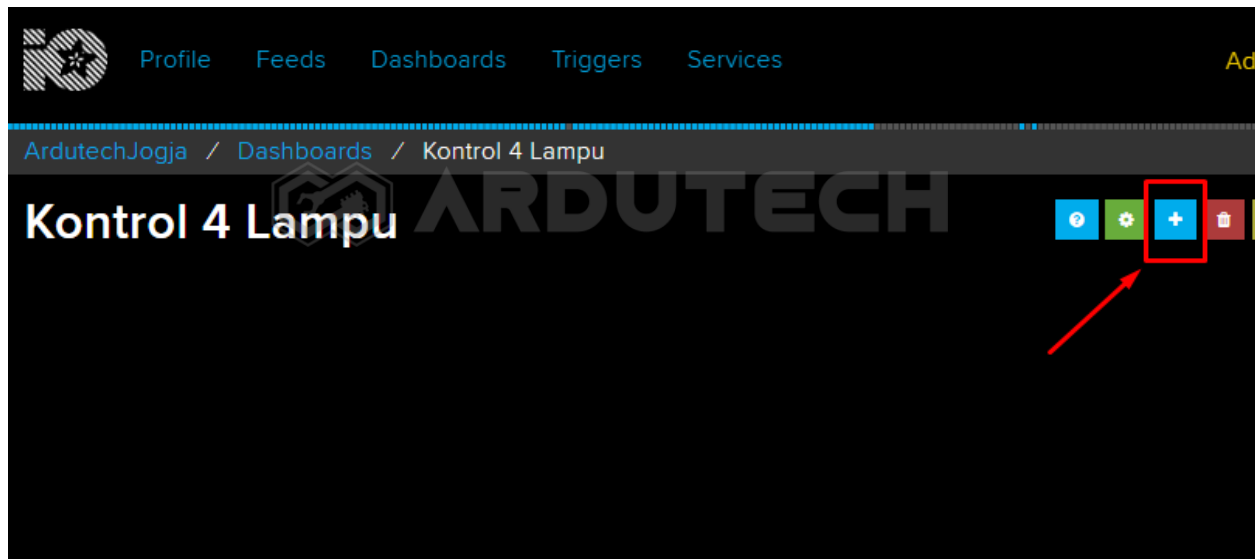
Name

Kontrol 4 Lampu

Description

Cancel Create

Setelah klik tombol **"Create"** , akan muncul dashboard baru dengan nama **"Kontrol 4 Lampu"**.

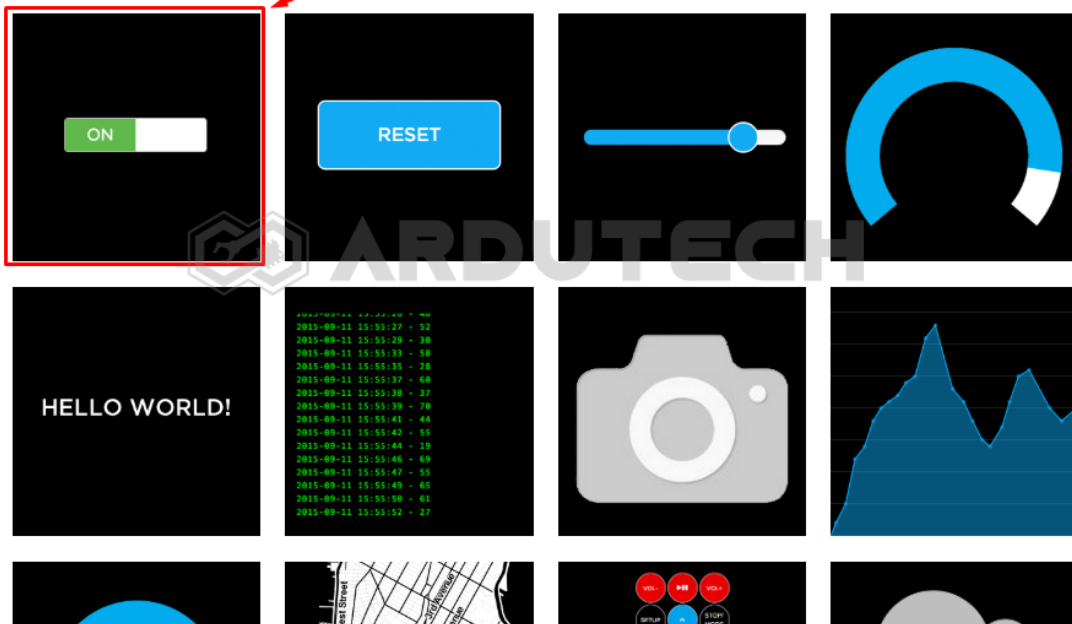


Klik **"Kontrol 4 Lampu"** kemudian pilih tombol **"+"** untuk membuat blok baru.

Setelah tombol **"+"** di-klik akan muncul tampilan bermacam 'widget'. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Toggle** (pojok kiri atas).

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'feed'. Ini semacam "Topik" dalam komunikasi MQTT. Jadi supaya dikenali oleh 'subscriber' (penerima perintah) maka 'publisher' harus menentukan 'topik' nya. Silakan tulis "LED" kemudian klik tombol "Create".

Choose feed

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
--------------	------------	----------

Setelah tombol "Create" diklik dan selanjutnya "Next Step" sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Kontrol Lampu 1

Button On Text

ON1

Button Off Text

OFF1

Block Preview

Kontrol Lampu 1

ON1

OFF1

Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

45

Published Value

0 bytes

Previous step

Create block

Isi pada kolom yang tersedia :

- Block Title : Kontrol Lampu 1 (opsional)
- Button On Text : ON1
- Button Off Text : OFF1

Klik tombol **"Create block"**

Tambahkan komponen Toggle lagi, kemudian pilih Feed : **LED**.

Choose feed

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☒ LED	OFF	12 minutes
☐ Light	0	about 1 hour
☐ photocell	561	17 minutes

Pilih LED kemudian klik **"Next Step"** muncul Block settings :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Kontrol Lampu 2

Button On Text

ON2

Button Off Text

OFF2

Block Preview

Kontrol Lampu 2

ON2

OFF2

Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

45

Published Value

0 bytes

Isi pada kolom yang tersedia :

- Block Title : Kontrol Lampu 2 (opsional)
- Button On Text : ON2
- Button Off Text : OFF2

Klik tombol **“Create block”**

Tambahkan komponen Toggle lagi, kemudian pilih Feed : **LED**.

Choose feed ✕

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☒ LED	OFF	12 minutes
☐ Light	0	about 1 hour
☐ photocell	561	17 minutes

Pilih LED kemudian klik **“Next Step”** muncul Block settings :

Block settings

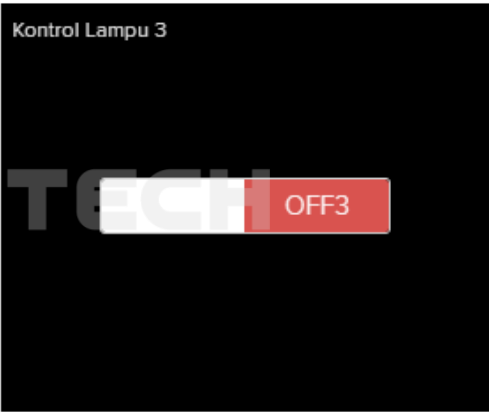
In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Button On Text

Button Off Text

Block Preview



Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

Published Value

0 bytes

[< Previous step](#)
[Create block](#)

Isi pada kolom yang tersedia :

- Block Title : Kontrol Lampu 3 (opsional)
- Button On Text : ON3
- Button Off Text : OFF3

Klik tombol **"Create block"**

Tambahkan komponen Toggle lagi, kemudian pilih Feed : **LED**.

Choose feed

Toggle: A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
☒ LED	OFF	12 minutes
☐ Light	0	about 1 hour
☐ photocell	561	17 minutes

< Previous step Next step >

Pilih LED kemudian klik **"Next Step"** muncul Block settings :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Kontrol Lampu 4

Button On Text

ON4

Button Off Text

OFF4

Block Preview

Kontrol Lampu 4

OFF4

Toggle A toggle button is useful if you have an ON or OFF type of state. You can configure what values are sent on press and release.

Test Value

Published Value

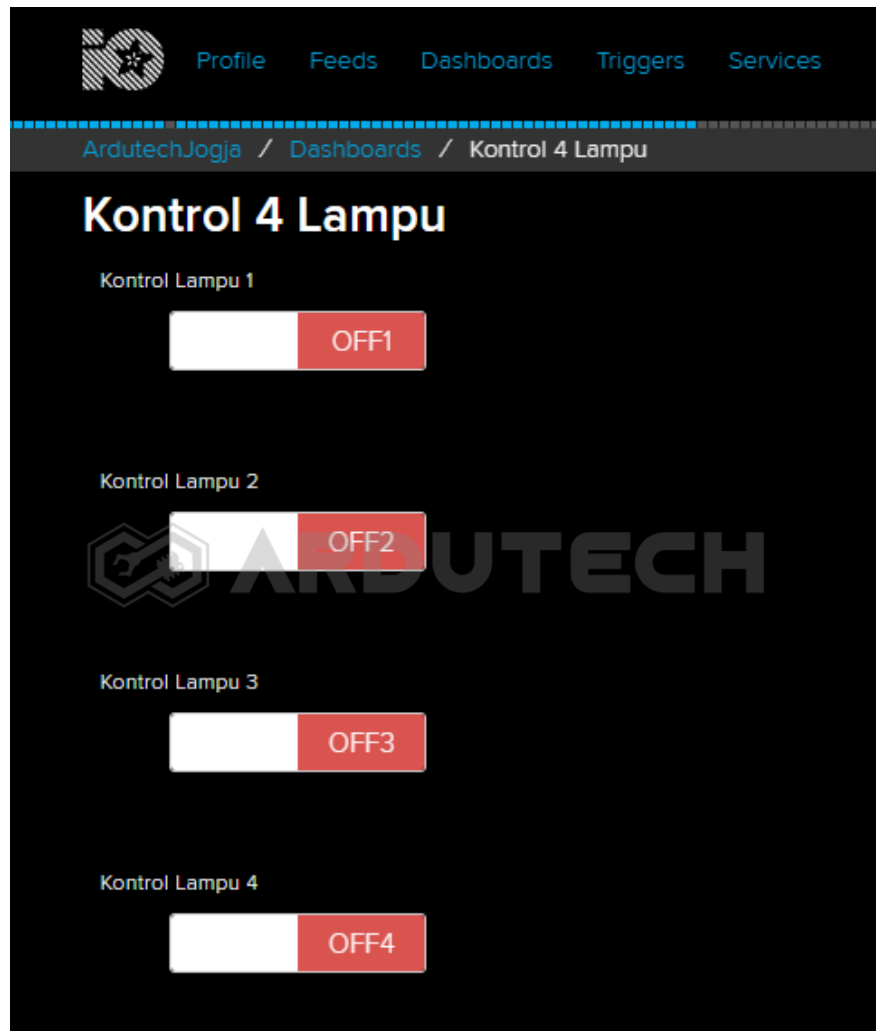
0 bytes

< Previous step Create block

Isi pada kolom yang tersedia :

- Block Title : Kontrol Lampu 4 (opsional)
- Button On Text : ON4
- Button Off Text : OFF4

Klik tombol **“Create block”**. Selesai pembuatan antarmuka di Adafruit.IO.



Selanjutnya kita perlu kode (token) untuk mengisi variabel username dan active key di program Arduino yang akan dibuat. Klik bagian **“Adafruit IO Key”** di pojok kanan atas.

YOUR ADAFRUIT IO KEY

Your Adafruit IO Key should be kept in a safe place and treated with the same care as your Adafruit username and password. People who have access to your Adafruit IO Key can view all of your data, create new feeds for your account, and manipulate your active feeds.

If you need to regenerate a new Adafruit IO Key, all of your existing programs and scripts will need to be manually changed to the new key.



Username

ArdutechJogja

Active Key

ruu_wUXJ670DJNEIBQWERTY567MNB

REGENERATE KEY

[Hide Code Samples](#)

Arduino

```
#define IO_USERNAME "ArdutechJogja"
#define IO_KEY      "ruu_wUXJ670DJNEIBQWERTY567MNB"
```

Perhatikan untuk **Username** dan **Active Key** ini penting, karena nanti dipakai ketika pemrograman dengan Arduino IDE. Catat/simpan kode tersebut.

Nah untuk pembuatan antarmuka pada Adafruit sudah selesai, sekarang kita siapkan program Arduino IDE-nya.

Program/Source Code

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h (sudah include ketika instalasi ESP8266)
- Adafruit_MQTT.h
- Adafruit_MQTT_Client.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/******
 * Project Kontrol 4 Lampu dg Adafruit IO
 * Board   : NodeMCU ESP8266 V3
 * Input    : MQTT (tombol) Adafruit IO
 * Output   : Relay (Lampu)
 * 99 Proyek IoT
 * www.ardutech.com
 *****/

#include <ESP8266WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
```

```
//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
#define WLAN_SSID      "ArdutechWiFi" // Nama Hotspot/WiFi
#define WLAN_PASS      "firmas20"     // Password
#define AIO_SERVER      "io.adafruit.com"
#define AIO_SERVERPORT  1883           // use 8883 for SSL
//---GANTI SESUAI DENGAN USER NAME ADAFRUIT ANDA
#define AIO_USERNAME    "ArdutechJogja"
//---GANTI SESUAI DENGAN IO KEY ADAFRUIT ANDA
#define AIO_KEY          "aio_mAXJ90DJnEIBVj0IDmG6BIMZqMAW"

WiFiClient client;
Adafruit_MQTT_Client    mqtt(&client,    AIO_SERVER,    AIO_SERVERPORT,
AIO_USERNAME, AIO_KEY);
Adafruit_MQTT_Publish   photocell       =    Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/photocell");
Adafruit_MQTT_Subscribe onoffbutton     =    Adafruit_MQTT_Subscribe(&mqtt,
AIO_USERNAME "/feeds/LED");

/***** Sketch *****/
/***** Code *****/

void MQTT_connect();

//=====

void setup() {
  Serial.begin(9600);
  pinMode(D5,OUTPUT);
  pinMode(D6,OUTPUT);
  pinMode(D7,OUTPUT);
  pinMode(D8,OUTPUT);
  delay(10);
  Serial.println(); Serial.println();
  Serial.print("Connecting to... ");
  Serial.println(WLAN_SSID);
```

```
WiFi.begin(WLAN_SSID, WLAN_PASS);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println();
Serial.println("WiFi connected");
Serial.println("IP address: "); Serial.println(WiFi.localIP());
mqtt.subscribe(&onoffbutton);
}
uint32_t x=0;
//=====
void loop() {
    MQTT_connect();
    Adafruit_MQTT_Subscribe *subscription;
    while ((subscription = mqtt.readSubscription(5000))) {
        if (subscription == &onoffbutton) {
            Serial.print(F("Got: "));
            Serial.println((char *)onoffbutton.lastread);
            if(strcmp((char*)onoffbutton.lastread,"ON1")==0){
                digitalWrite(D5,HIGH);
            }
            else if(strcmp((char*)onoffbutton.lastread,"OFF1")==0){
                digitalWrite(D5,LOW);
            }
            else if(strcmp((char*)onoffbutton.lastread,"ON2")==0){
                digitalWrite(D6,HIGH);
            }
            else if(strcmp((char*)onoffbutton.lastread,"OFF2")==0){
                digitalWrite(D6,LOW);
            }
        }
    }
}
```

```
        else if(strcmp((char*)onoffbutton.lastread,"ON3")==0){
            digitalWrite(D7,HIGH);
        }
        else if(strcmp((char*)onoffbutton.lastread,"OFF3")==0){
            digitalWrite(D7,LOW);
        }
        else if(strcmp((char*)onoffbutton.lastread,"ON4")==0){
            digitalWrite(D8,HIGH);
        }
        else if(strcmp((char*)onoffbutton.lastread,"OFF4")==0){
            digitalWrite(D8,LOW);
        }
    }
}

//=====
void MQTT_connect() {
    int8_t ret;
    // Stop if already connected.
    if (mqtt.connected()) {
        return;
    }
    Serial.print("Connecting to MQTT... ");
    uint8_t retries = 3;
    while ((ret = mqtt.connect()) != 0) { // connect will return 0 for
connected
        Serial.println(mqtt.connectErrorString(ret));
        Serial.println("Retrying MQTT connection in 5 seconds...");
        mqtt.disconnect();
        delay(5000); // wait 5 seconds
        retries--;
        if (retries == 0) {
```



```
// basically die and wait for WDT to reset me
while (1);
}
}
Serial.println("MQTT Connected!");
}
```

PERHATIKAN !

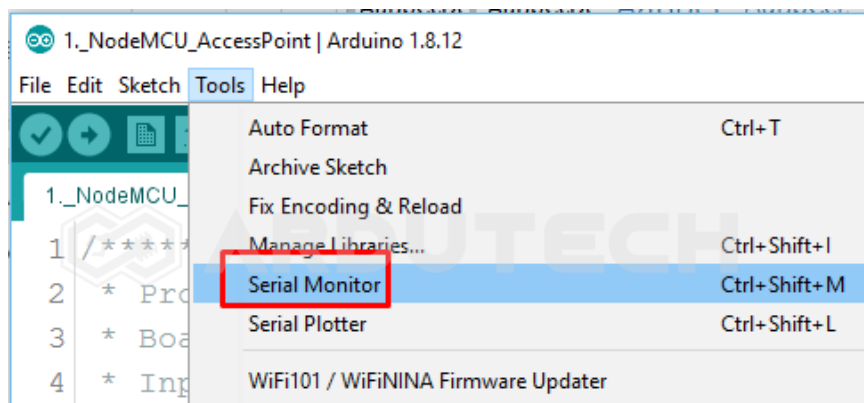
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **WLAN_SSID**
- Password jaringan WiFi/hotspot : **WLAN_PASS**
- Username Adafruit IO : **AIO_USERNAME**
- Active Key Adafruit IO : **AIO_KEY**

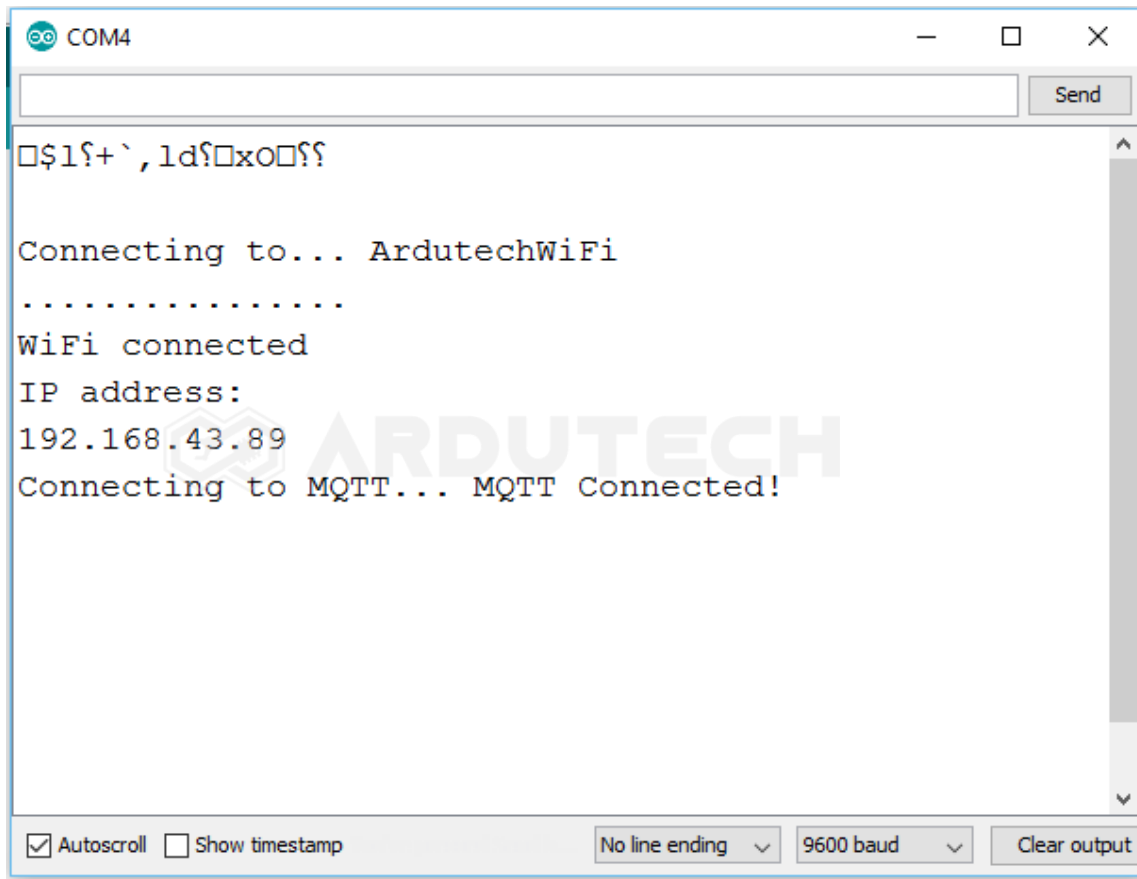
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

Jalannya Alat

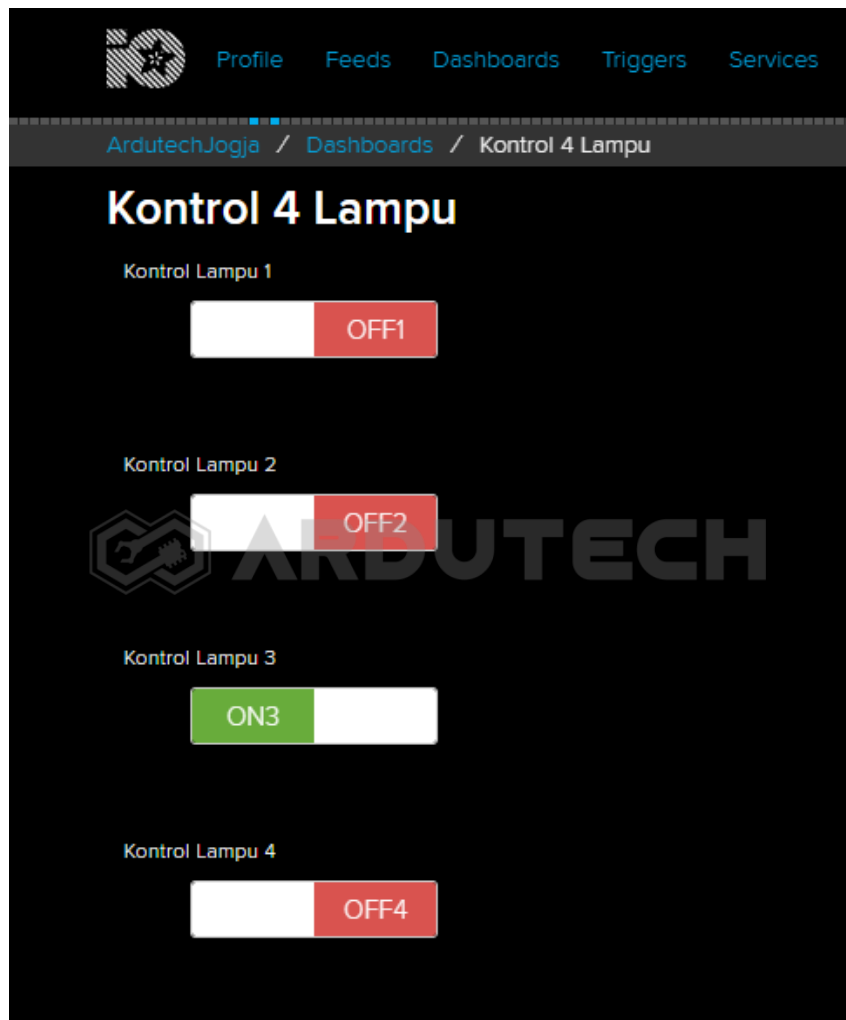
Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools → Serial Monitor** kemudian seting baudrate pada nilai 9600.



Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :



Setelah berhasil terhubung dengan jaringan WiFi sekarang kita kembali ke tampilan di Adafruit. Tekan tombol ON – OFF dan perhatikan Relay dan lampu yang terhubung dengan pin NodeMCU.



Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – “Sahabat Inovasi Anda”